Understanding the Implicit function theorem

Soham Saha

https://soham-saha.github.io

https://soham-saha.codeberg.page

This is an attempt at gaining "intuition", an extremely versatile, yet painfully error-prone tool in a mathematician's toolbox.

Let us start with a simple example first.

Q. You are given the unit 3-dimensional sphere $x^2 + y^2 + z^2 = 1$, and a plane $ax + by + cz = k$ such that the plane intersects the sphere in a circle C. Define π_X as the projection map from $\mathbb{R}^3$ to the X-axis. Given a point $p \in C$, what minimal constraints do you need to guarantee the existence of an open set U around p such that points in $\pi_X(U)$ uniquely correspond to points in $U \cap C$ (i.e., the map π_X from $U \cap C$ to $\pi_X(U)$ will be a bijection)?

A. Observe that whenever the plane is perpendicular to the X-axis, there can exist no such U.

When the plane is inclined with respect to the X-axis, there are exactly two points where an open set U as described above cannot exist, namely the points which have the minimum and maximum possible X-coordinate (in C) respectively.

For all other points apart from these two points, we can get such an open set U as specified in the question.

Let us analyze what is happening in the example above a bit further, setting up the scene for generalization.

Let us call the points in C as "good", and points outside C as "bad". We are testing a certain nice property of the intersection of the zero-locus of $x^2 + y^2 + z^2 - 1$ and $ax + by + cz - k$ at the point p.

Let us define $f := (x, y, z) \mapsto (x^2 + y^2 + z^2 - 1, ax + by + cz - k)$ as the "goodness" function; since a point q is good $\Leftrightarrow f(q) = 0 \Leftrightarrow \|f(q)\| = 0$, and *crucially*, points with lower $\|f(q)\|$ are "closer" to good points than points with higher $\|f(q)\|$.

The key observation now is that $\det\big(Df_{y,z}(p)\big)$ is the measure of how fast points around p (in $\{p_X\} \times \mathbb{R}^2$) go bad. Since p was good to start with, $\det\big(Df_{y,z}(p)\big)$ being 0 would imply a high concentration of good points nearby (either on the plane $\{p_X\} \times \mathbb{R}^2$ itself, or very close to that plane), suggesting that the local uniqueness property may not be satisfied.

Observe that this is only a "suggestion", in the sense that it is possible to have $\det\big(Df_{y,z}(p)\big) = 0$ and yet satisfy the locally unique property. Think of $f := (x, y, z) \mapsto (y^3 - x, z)$ at $(0, 0, 0)$.

So the determinant of $\det\big(Df_{y,z}(p)\big) = 0$ **doesn't** guarantee that the zero-locus is not locally like a graph at near p; it may or may not be so.

It *may* be the case that when $\det\big(Df_{y,z}(p)\big) = 0$ and yet the zero-locus is locally like a graph near p, the graph would not be of a C^1 function. However, I am not entirely sure about this.

On the other hand, $\det\big(Df_{y,z}(p)\big) \neq 0$ implies a rapid change in goodness around p in $\{p_X\} \times \mathbb{R}^2$, and guarantees that the local uniqueness property stated in the question is being satisfied. Therefore, we focus on the non-singularity of the rightmost square block in $Df(p)$, that is, the non-singularity of $\big(Df_{y,z}(p)\big)$.

Note that the "goodness" function f is just a book-keeping function for what constraints we are trying to solve for. It being C^1 tells us that the constraints we are dealing with are nice and smooth, and its rate of change locally around a point (i.e., the determinant of its derivative) describes the rate of change of "goodness" around that point.

Now, let us try to generalize these thoughts.

Q. Suppose we have $n+m$ variables, conveniently named $x_1, ..., x_n, y_1, ..., y_m$ (we shall abbreviate them as x, y), and m constraints on them, namely $f_1(x,y) = ... = f_m(x,y) = 0$. Assume that all f_i are continuously differentiable. Let us call $C \subseteq \mathbb{R}^{n+m}$ the set of points in $\mathbb{R}^{n+m}$ which satisfy the constraints. Define $\pi : \mathbb{R}^{n+m} \to \mathbb{R}^n$ as the projection map onto the first n coordinates.

Assume that we have (a,b) such that the above constraints are satisfied by $x = a, y = b$. What minimal conditions do we need to guarantee the existence of an open set U around (a,b) in $\mathbb{R}^{n+m}$ such that points in $U \cap C$ have their last m coordinates uniquely determined by their first n coordinates (i.e., the map π from $U \cap C$ to $\pi(U)$ will be a bijection)?

A. As before, we define the "goodness" function as $f = (f_1, ..., f_m)$. The local uniqueness property is implied by a high rate of change of "goodness" of points close enough to (a,b) in $\{a\} \times \mathbb{R}^m$; which in turn is implied by $\det\big(D_y f(a,b)\big) \neq 0$, where we are considering the partial Jacobian obtained by differentiating only along the last m coordinates.

So, when $\det\big(D_y f(a,b)\big)$ is non-zero (which corresponds to the right-most $m \times m$ submatrix of $Df(a,b)$ being non-singular), we can get an open neighborhood U around (a,b) such that for every $x \in \pi(U)$, we can uniquely solve for y such that (x,y) is satisfies the constraints.

Therefore, $\det\big(Df_y(a,b)\big) \neq 0$ is a sufficient condition for the existence of open set U as described above, but it is not a necessary condition in general.

Now, to put the final nail in the coffin, let us have a look at what Wikipedia gives us as intuition for the implicit function theorem:

> *Under a mild condition on the partial derivatives, the set of zeros of a system of equations is locally the graph of a function.*

Unpacking the Wikipedian sentiment,

> Given a system of equations $f_1(x,y) = ... = f_m(x,y) = 0$ on variables $x_1, ..., x_n, y_1, ..., y_m$, the set $C \subseteq \mathbb{R}^{n+m}$ of solutions to those constraints and $(p,q) \in C$,
> - The "mild condition on partial derivatives" is that the right-most $m \times m$ submatrix of $Df(p,q)$ is invertible.
> - If that is the case, then we get that there exists an open neighborhood U around (p,q) such that
>
> $$\exists g, U \cap C = \{(x, g(x)) : x \in \pi(U)\},$$
>
> therefore showing that C is locally the graph of g at the value p.

A cleaner formulation, with a few added details would be:

Implicit function theorem

Given an open set $A \subseteq \mathbb{R}^{n+m}$, a C^1 function $f : A \to \mathbb{R}^m$, points $a \in \mathbb{R}^n, b \in \mathbb{R}^m$ such that $f(a,b) = 0$, and the partial Jacobian $D_y f(a,b)$ invertible, there exist open sets $U \ni a, V \ni b$ (with $U \times V \subseteq A$) and a C^1 function $g : U \to V$ such that

$$\{(x,y) \in U \times V : f(x,y) = 0\} = \{(x, g(x)) : x \in U\}$$

Moreover, g satisfies

$$\forall x \in U, Dg(x) = -\big(D_y f(x, g(x))\big)^{-1} \circ D_x f(x, g(x))$$

Therefore, given m constraints on the $n+m$ variables $x_1, ..., x_n, y_1, ..., y_m$, g locally solves for y in terms of x around $x = a, y = b$.

This is the formalization which we shall be using in the next sections.

The key thing to remember about the implicit function theorem is that it allows us to locally solve for y in terms of x given $f(x,y) = 0$.

Though the implicit function theorem is much more general that the inverse function theorem, we can give a nice proof of the implicit function theorem using the inverse function theorem.

First, let us state the inverse function theorem :

Inverse function theorem

Given an open set $A \subseteq \mathbb{R}^k$, a C^1 function $f : A \to \mathbb{R}^k$ and a point $p \in A$ such that $Df(p)$ is non-singular, there exists an open set $U \subseteq A$ which contains p, such that:

- $\forall x \in U, Df(x)$ is non-singular.
- $f(U)$ is open in $\mathbb{R}^k$.
- f is a C^1-diffeomorphism from U to $f(U)$. That is, $f|_U$ is a bijection from U to $f(U)$ with a continuously differentiable inverse.

Moreover, $g := (f|_U)^{-1}$ satisfies

$$\forall y \in f(U), Dg(y) = (Df(g(y)))^{-1},$$

or equivalently,

$$\forall x \in U, Dg(f(x)) = (Df(x))^{-1}$$

Now we shall prove the implicit function theorem, referring to the symbols in the statement of the implicit function theorem above.

Observe that

$$f(x,y) = 0 \Leftrightarrow F(x,y) = (x,0) \Leftrightarrow (x,y) \in F^{-1}(x,0),$$

where $F := (x,y) \mapsto (x, f(x,y))$.

Therefore, if we can locally invert F around (a,b), then we can extract g as

$$g := x \mapsto G_y(x,0) = \pi \circ G \circ i$$

where $\pi := (x, y) \mapsto y$, G is the local inverse of F, and $i := x \mapsto (x, 0)$.

The inverse function theorem gives us local invertibility of F, and we get the derivative of g by

$$\begin{aligned}
Dg(x) &= D(\pi \circ G \circ i)(x) \\
&= D\pi((G \circ i)x) \circ DG(i(x)) \circ Di(x) \\
&= \pi \circ DG(x, 0) \circ i \\
&= \pi \circ (DF(x, g(x)))^{-1} \circ i \\
&= \pi \circ \begin{bmatrix} I_n & 0 \\ D_x f(x, g(x)) & D_y f(x, g(x)) \end{bmatrix}^{-1} \circ i \\
&= -\big(D_y f(x, g(x))\big)^{-1} \circ D_x f(x, g(x))
\end{aligned}$$

Once we have the implicit function theorem, we can recover the inverse function theorem easily.

We want to locally invert $f : A \to \mathbb{R}^k$ around $p \in A$. So, we want to locally solve for y in terms of x in $f(y) = x \Leftrightarrow f(y) - x = 0$. Applying the implicit function theorem to $(x, y) \mapsto f(y) - x$ at the point $(f(p), p)$ gives us what we want.

References:

- https://en.wikipedia.org/wiki/Implicit_function_theorem : Mainly the introduction, *Definition* and the *Statement of Theorem* sections.
- https://en.wikipedia.org/wiki/Inverse_function_theorem#Implicit_function_theorem